ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

$\dot{x}_1 = x_2$ and $\dot{x}_2 = (\tau - b\, x_2 + mg\ell \sin\theta)/J$. At equilibrium:

$$_0 = -mg\ell sin\theta_e,$$

an equilibrium exists for $|\tau_0|$ less than or equal to 2.4525 N m

Q2(b) [2 marks]

$$A = [[0, 1], [-61.3125, -2.5]], B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is $K_p(\theta_d - \theta)$, which must balance $mg\ell \sin\theta$.

Q2(d) [2 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

Tustin uses $s = (2/T)(z-1)/(z+1)$.

Q3(c) [4 marks]

$A - LC = [[-\ell_1, 1], [-\ell_2, -2]]$. I obtain $p(s) = s^2 + (\ell_1 + 2)s + (\ell_1 + \ell_2)$. Matching $(s+10)^2$ gives $\ell_1 = 18$ and $\ell_2 = 82$.

Q3(d) [2 marks]

The observer is simpler because it only smooths the measured velocity. The filter is safer when the model is poor.

Question 4 [10 marks]

Q4(a) [2 marks]

The closed-loop poles are -2±2j. I use Nff=c/b=6 for unity DC gain.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) [4 marks]

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.